FOUNDER EDITION V1.0

AI-Native EHS Starter Kit

Practical prompts, workflows, guardrails, and career tools for EHS, HSE, OSH, and safety leaders using AI responsibly.

WHAT THIS KIT HELPS YOU DO

Use AI as a practical assistant for drafting, review, communication, and learning while keeping safety accountability, verification, and final decisions with competent people.

50	**5**	**30 days**
copy-ready EHS prompts	AI-assisted workflows	career upgrade plan

Quick start

1. Choose one real task: inspection report, incident learning, JHA review, toolbox talk, or monthly EHS briefing.
2. Remove confidential, personal, legal-sensitive, medical, and investigation-sensitive details before using any public AI tool.
3. Use the SAFER prompt formula: Situation, Aim, Facts, Evidence constraints, Review requirements.
4. Ask AI for a draft, then verify it against laws, company standards, permits, OEM guidance, and actual field conditions.
5. Save the final prompt and output as a reusable workflow only after a competent person approves it.

How to read this guide

This is not legal advice, engineering approval, or a replacement for competent safety management. It is a practical starter kit for better thinking, faster drafts, clearer communication, and more disciplined verification.

1. Responsible AI guardrails for EHS work

CORE RULE

AI can assist the EHS professional. It cannot own safety accountability, approve high-risk work, replace field verification, or make final legal or operational decisions.

Use AI for

- Drafting first versions of reports, checklists, training scripts, and management summaries.
- Finding gaps, unclear wording, weak controls, and missing verification evidence.
- Creating structured questions for inspections, investigations, reviews, and coaching conversations.
- Translating technical safety content into plain language for different audiences.

Do not use AI for

- Uploading personal data, confidential company data, sensitive investigation records, or protected legal documents into public tools.
- Approving high-risk work such as LOTO, confined space, critical lifting, hot work, electrical isolation, pressure systems, or emergency response.
- Replacing qualified engineering review, regulatory interpretation, medical advice, or legal counsel.
- Creating fake evidence, hiding uncertainty, or presenting unverified output as a final professional decision.

Verification checklist

1. What facts did AI use, and which facts are still missing?
2. Does the output match local laws, company standards, permits, and site procedures?
3. Does it identify critical controls and how those controls will be verified in the field?
4. Does it avoid blame, exaggeration, unsupported claims, and confidential details?
5. Who is the competent human owner for final approval?

2. The SAFER prompt formula

Element	What to include
Situation	Workplace, task, audience, risk profile, and decision context.
Aim	What output you need and how it will be used.
Facts	Only verified, sanitized facts and relevant constraints.
Evidence	Standards, procedures, permits, data, or documents the output must respect.
Review	Ask AI to list assumptions, gaps, verification needs, and human approval points.

Reusable starter prompt

COPY THIS

Act as a practical EHS advisor. I will give you sanitized workplace information. Help me produce a draft, but separate facts from assumptions, flag missing information, identify verification steps, and remind me where competent human approval is required.

3. 50 EHS AI prompts

Use these prompts as starting points. Replace the generic task description with your sanitized context, then verify every output before use.

Inspection and Observation

1. **Convert rough notes into a site observation:** Act as an EHS manager. Convert these rough site notes into a professional observation report with: finding, likely risk, immediate action, recommended permanent control, owner, due date, and verification evidence. Keep assumptions separate from facts.
2. **Turn photos into inspection questions:** Based on the described work scene, create 12 inspection questions for a supervisor walkdown. Group them by people, plant, process, environment, and emergency readiness. Highlight any questions that require direct field verification.
3. **Prioritize open actions:** Review this list of open EHS actions and classify each as critical, high, medium, or low. Use severity, exposure, legal duty, and repeat history. Return a concise action plan for the top five items.
4. **Create a weekly inspection route:** Design a 60-minute EHS inspection route for this workplace. Include checkpoints, questions to ask workers, documents to sample, and evidence to capture. Make the route practical for a busy operations manager.
5. **Rewrite a finding without blame:** Rewrite this inspection finding in neutral, professional language. Avoid blame, keep the risk clear, and include one immediate control and one system-level follow-up.
6. **Spot weak corrective actions:** Review these corrective actions and identify which are weak, unclear, or unlikely to prevent recurrence. Rewrite them using action-owner-deadline-evidence format.
7. **Build a supervisor checklist:** Create a one-page supervisor checklist for this activity. Include pre-start checks, critical controls, stop-work triggers, and close-out checks.
8. **Summarize trend signals:** Analyze these inspection findings and identify recurring themes, weak controls, and leading indicators. Produce a management summary with three recommended interventions.
9. **Prepare a contractor walkdown script:** Create a contractor safety walkdown script for a project manager. Include opening questions, field checks, worker engagement prompts, and a closing commitment conversation.
10. **Create an evidence request:** Draft a concise evidence request for closing this EHS action. Specify acceptable evidence, unacceptable evidence, and what the reviewer should verify before closure.

Incident Learning and Investigation

11. **Structure an incident narrative:** Convert these incident notes into a clear timeline. Separate confirmed facts, missing information, witness statements, immediate causes, system factors, and questions for follow-up.
12. **Generate interview questions:** Create non-leading interview questions for an incident investigation. Include questions for the injured person, supervisor, witnesses, maintenance, and management.

13. **Identify possible system factors:** Review this incident summary and list possible system factors, including planning, supervision, competence, design, workload, communication, procedures, and management of change.
14. **Turn lessons learned into action:** Turn this incident lesson learned into a practical action plan. Include immediate containment, permanent controls, communication, training, verification, and owner accountability.
15. **Draft a learning alert:** Draft a one-page safety learning alert for frontline teams. Use plain language, avoid confidential details, explain what happened, why it matters, and what everyone should do differently.
16. **Challenge a weak root cause:** Review this proposed root cause statement. Identify if it blames human error too quickly, misses system conditions, or lacks evidence. Suggest a stronger root cause statement.
17. **Create a recurrence-prevention check:** Create a recurrence-prevention verification checklist for this incident. Include field evidence, training evidence, supervisor behavior, engineering controls, and management review.
18. **Build a five-whys conversation:** Create a five-whys facilitation guide for this incident. Make each why evidence-based and include prompts to avoid stopping at worker behavior.
19. **Summarize for executives:** Summarize this incident investigation for executives in five bullets: what happened, actual/potential consequence, system weakness, immediate action, and decision needed.
20. **Create a legal-sensitive version:** Rewrite this incident communication for broad learning without admitting liability, naming individuals, or sharing sensitive personal information. Keep the safety lesson specific and useful.

Risk Assessment and JHA/JSA

21. **Review a JHA for missing hazards:** Review this JHA/JSA and identify missing hazards, weak controls, unclear steps, and verification gaps. Return a revised version with critical controls highlighted.
22. **Create a task risk prompt:** For this task, create a pre-task risk assessment with steps, hazards, initial risk, controls, residual risk, required permits, stop-work triggers, and supervisor verification.
23. **Stress-test controls:** Stress-test these risk controls. For each control, explain how it could fail, what evidence proves it is working, and what backup control is needed for high-risk work.
24. **Compare control options:** Compare these control options using hierarchy of controls, practicality, cost, worker acceptance, maintainability, and residual risk. Recommend the strongest balanced option.
25. **Create a dynamic risk brief:** Create a 5-minute dynamic risk assessment brief for field supervisors. Include changing conditions, interfaces, energy sources, line-of-fire, simultaneous operations, and escalation triggers.
26. **Build a permit review guide:** Create a permit-to-work review guide for this activity. Include key permit checks, isolation evidence, competence checks, emergency arrangements, and handback requirements.
27. **Identify critical controls:** Identify the critical controls for this high-risk activity. For each one, define the owner, performance standard, verification method, and frequency.
28. **Translate technical risk for leaders:** Translate this technical risk assessment into a leadership briefing. Focus on decision points, unacceptable conditions, required resources, and what leaders must visibly verify.

29. **Create a risk bowtie starter:** Create a bowtie analysis starter for this hazard. Include top event, threats, preventive controls, consequences, mitigating controls, escalation factors, and assurance questions.
30. **Review management of change:** Review this planned change from an EHS perspective. Identify new hazards, affected procedures, training needs, communication needs, approvals, and post-change verification.

Training and Communication

31. **Create a toolbox talk:** Create a 10-minute toolbox talk for this topic. Include a hook, three key messages, one short scenario, worker discussion questions, and a supervisor close-out commitment.
32. **Turn policy into plain language:** Rewrite this EHS policy into plain frontline language. Keep the legal intent, remove jargon, and include what workers must do, what supervisors must check, and when to stop work.
33. **Create a microlearning script:** Create a 3-minute microlearning video script with voiceover, scene notes, on-screen text, and one knowledge check question. Keep it realistic for EHS professionals.
34. **Build quiz questions:** Create 10 scenario-based quiz questions for this training topic. Include correct answer, explanation, and what wrong answers reveal about misunderstanding.
35. **Design a leadership safety moment:** Create a two-minute leadership safety moment for senior managers. Make it credible, operational, and focused on what leaders should ask and verify.
36. **Create multilingual briefing notes:** Convert this safety message into simple briefing notes for a diverse workforce. Use short sentences, avoid idioms, and include picture or demonstration suggestions.
37. **Plan refresher training:** Create a refresher training plan for this recurring issue. Include audience, learning objective, field practice, supervisor reinforcement, and evidence of effectiveness.
38. **Create behavior observation coaching:** Create a coaching guide for a safety behavior observation. Include what to observe, how to open the conversation, how to give feedback, and how to record learning.
39. **Write a campaign message:** Create a one-month safety campaign message plan for this risk. Include weekly themes, poster headlines, toolbox questions, and leader talking points.
40. **Turn data into a story:** Turn these EHS metrics into a short story for employees. Explain the trend, why it matters, what is improving, and the one behavior or control to focus on next week.

Management Reporting and Career Growth

41. **Create an executive dashboard narrative:** Write an executive EHS dashboard narrative from these metrics. Highlight leading indicators, risk exposure, overdue actions, weak signals, and decisions needed from leadership.
42. **Prepare a board-level risk note:** Prepare a board-level note on this EHS risk. Include current exposure, regulatory concern, operational impact, control maturity, investment need, and next 90-day actions.
43. **Draft a monthly EHS review:** Draft a monthly EHS review agenda that focuses on learning, critical controls, contractor risk, audit actions, training effectiveness, and leader accountability.
44. **Create an AI use-case shortlist:** Create a shortlist of responsible AI use cases for an EHS function. Rank by value, feasibility, data sensitivity, operational risk, and ease of human verification.

45. **Write LinkedIn positioning:** Write five LinkedIn headline options for an EHS professional moving toward AI-enabled safety leadership. Keep the tone credible, practical, and not exaggerated.
46. **Upgrade CV achievement bullets:** Rewrite these CV bullets to show impact, leadership, risk reduction, cross-functional influence, and AI-readiness. Keep claims truthful and measurable.
47. **Create a 30-day learning plan:** Create a 30-day AI learning plan for an EHS professional. Include weekly goals, practice tasks, prompt experiments, verification habits, and a portfolio artifact.
48. **Prepare interview stories:** Create STAR interview stories from these EHS achievements. Emphasize leadership, stakeholder alignment, risk judgment, and practical innovation.
49. **Design a personal AI policy:** Create a personal responsible AI policy for an EHS professional. Include data boundaries, verification rules, disclosure principles, and prohibited use cases.
50. **Create a consulting offer outline:** Help me outline a small consulting or training offer based on my EHS experience. Include target buyer, pain point, outcome, modules, price logic, and proof needed.

4. Workflow templates

A workflow is more valuable than a single prompt. Use the table below to turn repeatable EHS tasks into controlled AI-assisted routines.

Workflow	How to use it	Human verification
Inspection to action closure	Capture field facts, classify risk, draft neutral finding, propose corrective action, and verify closure evidence.	AI drafts the report and action wording; the EHS owner verifies facts, legal duties, and field evidence.
Incident learning alert	Build a timeline, separate facts from assumptions, identify system factors, and produce a shareable learning alert.	AI structures learning; the investigator validates evidence, confidentiality, legal sensitivity, and root-cause quality.
JHA/JSA critical-control review	Review task steps, hazards, critical controls, stop-work triggers, and supervision checks before work starts.	AI challenges gaps; the supervisor confirms actual site conditions and control availability.
Toolbox talk microlearning	Convert a risk topic into a short briefing, scenario, discussion questions, and knowledge check.	AI creates a draft; the trainer adapts language, examples, and local procedure references.
Weekly EHS leadership brief	Summarize indicators, weak signals, overdue actions, contractor issues, and decisions needed.	AI creates a concise narrative; leaders validate numbers, context, and commitments.

Workflow operating rhythm

1. Create a controlled prompt template for each recurring EHS task.
2. Define what data can and cannot be pasted into the AI tool.
3. Require the AI output to list assumptions, missing data, and human verification steps.
4. Review output quality with a competent EHS professional before sharing.
5. Improve the prompt after each use and store the approved version in a team knowledge base.

5. Career upgrade pack

Positioning lines

- EHS leader helping high-risk organisations adopt AI responsibly without weakening safety governance.
- Safety professional combining operational risk control, frontline engagement, and practical AI workflow design.
- AI-native EHS practitioner focused on incident learning, training innovation, and critical-control assurance.

CV bullet examples

- Built AI-assisted EHS reporting workflow that reduced drafting time while maintaining human verification and governance controls.

- Translated complex risk data into executive-ready insights, enabling faster prioritisation of critical controls and overdue actions.
- Developed practical training scripts and scenario-based learning assets to improve supervisor safety conversations.

30-day AI-native EHS practice plan

Period	Action
Week 1	Learn prompting basics, create your personal responsible AI rules, and rewrite one EHS report with AI assistance.
Week 2	Build prompts for inspection findings, toolbox talks, JHA review, and incident learning alerts.
Week 3	Create one workflow template, test it on sanitized examples, and collect reviewer feedback.
Week 4	Publish a portfolio artifact: a sample checklist, training script, dashboard narrative, or AI safety reflection.

6. Mini practice case

SCENARIO

A contractor maintenance team had a near miss when a worker entered the line-of-fire during equipment repositioning. No injury occurred. The supervisor wants a learning alert, a toolbox talk, and stronger pre-task risk checks.

Try this sequence

1. Use prompt 11 to structure the incident timeline from sanitized facts.
2. Use prompt 13 to identify possible system factors beyond worker behavior.
3. Use prompt 25 to create a dynamic risk brief for supervisors.
4. Use prompt 31 to create a 10-minute toolbox talk.
5. Use the verification checklist before sharing any output.

What good output should include

- A clear line-of-fire explanation that workers can understand.
- A practical pre-task discussion for equipment movement and exclusion zones.
- A supervisor verification action, not only a worker instruction.
- A record of missing facts and assumptions that still need investigation.

Next step with Alvin

If you want to move from individual AI use to an organisational AI safety roadmap, start with a corporate discovery call or the AI Safety Transformation Diagnostic.

Portfolio and booking: https://vibe-portfolio-dny.pages.dev/

LinkedIn: https://www.linkedin.com/in/ir-bo-alvin-liao-2b237b95/

Advisory note: always adapt AI-assisted outputs to your jurisdiction, regulatory duties, internal governance, site risk profile, and competence requirements.